Name : R.Gowsika

Date : 27-10-2026

Tobic : Wireshark

What is wireshark?

Wireshark is a free, open-source tool that captures and displays the data packets moving across a network, letting you see exactly what devices are sending and receiving. It breaks each packet into its layers and decodes thousands of protocols, so network engineers use it to troubleshoot problems, security analysts use it to spot threats like malware traffic, and students use it to learn how networks work. It can't read encrypted (TLS) content without keys, and it should only be used on networks you own or have permission to monitor.

Step 1: Install Wireshark

1. Download Wireshark from the official website.
2. Run the installer and accept the defaults, including Npcap (Windows).
3. Complete the installation.

Step 2: Open Wireshark and Start the Capture

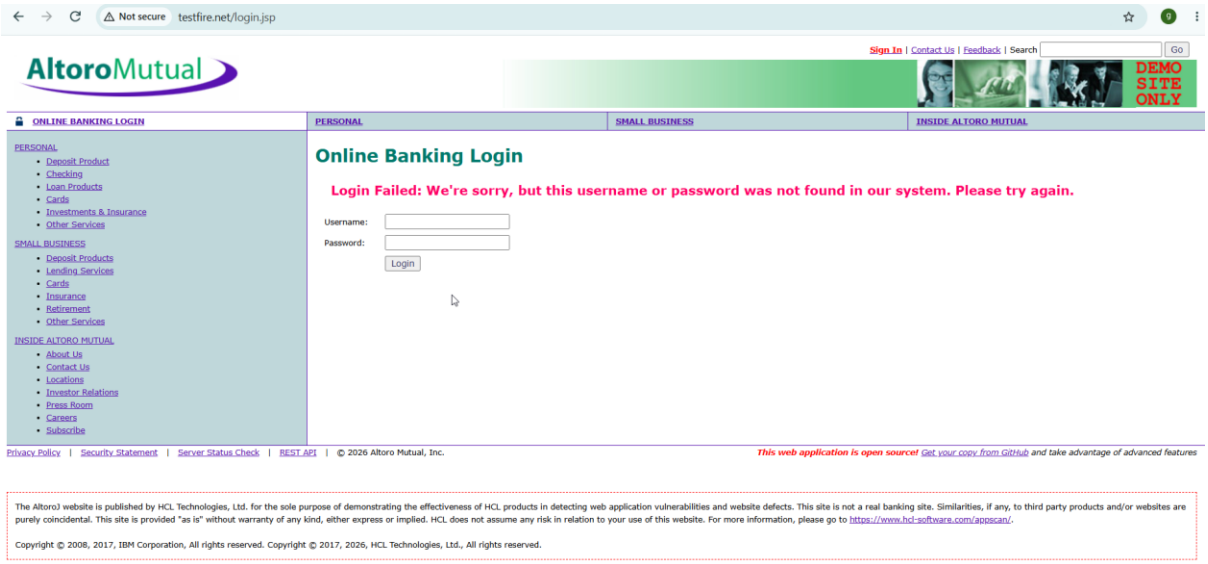
1. Launch Wireshark.
2. Select the active network interface (Wi-Fi or Ethernet), identified by the moving activity graph.
3. Click the blue shark-fin icon to start capturing. Start the capture before browsing so no traffic is missed

Step 3: Open Google

1. Open a web browser and visit www.google.com.
2. This generates DNS and HTTPS traffic, which serves as a comparison. Google uses encryption, so its content is not readable.

Step 4: Open testfire.net

1. In the browser, visit http://testfire.net.
2. Click **Sign In**.
3. Enter the demo credentials (for example, username: admin, password: admin) and click **Login**.
4. Login was wrong. The wireshark will record the process.

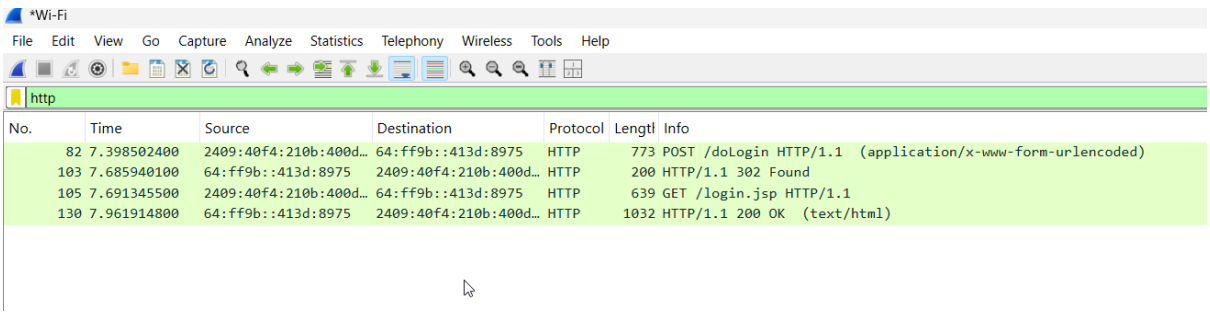


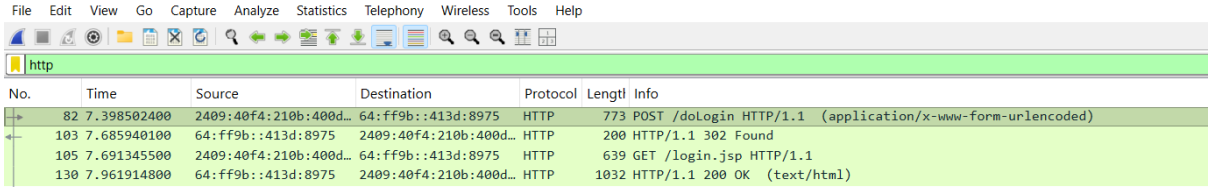
Step 5: Stop the Capture

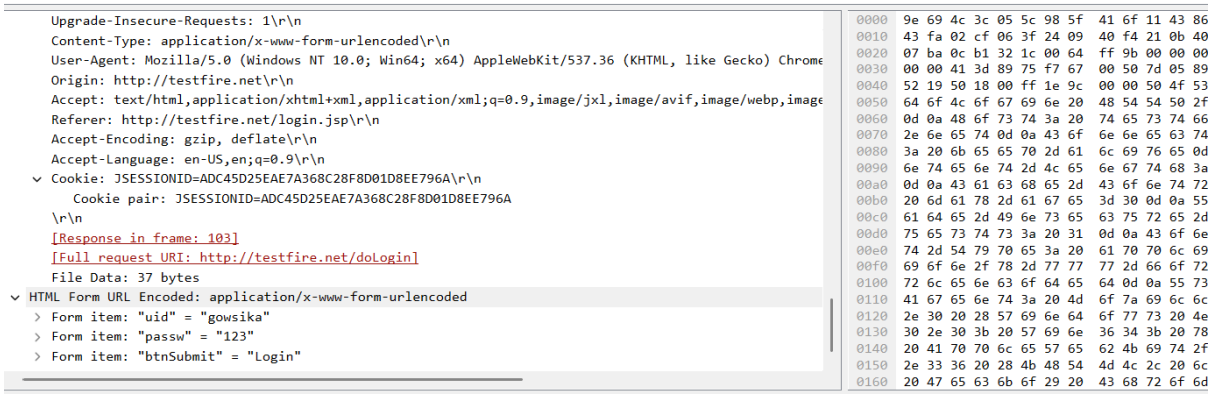
1. Return to Wireshark and click the red Stop button.
2. Save the capture using File > Save As (format: .pcapng).

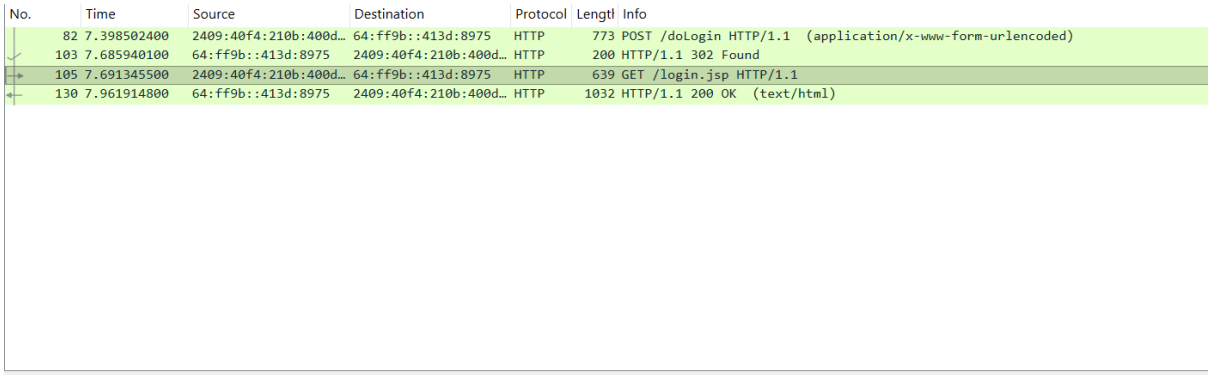
Step 6: Analyze the Captured Traffic

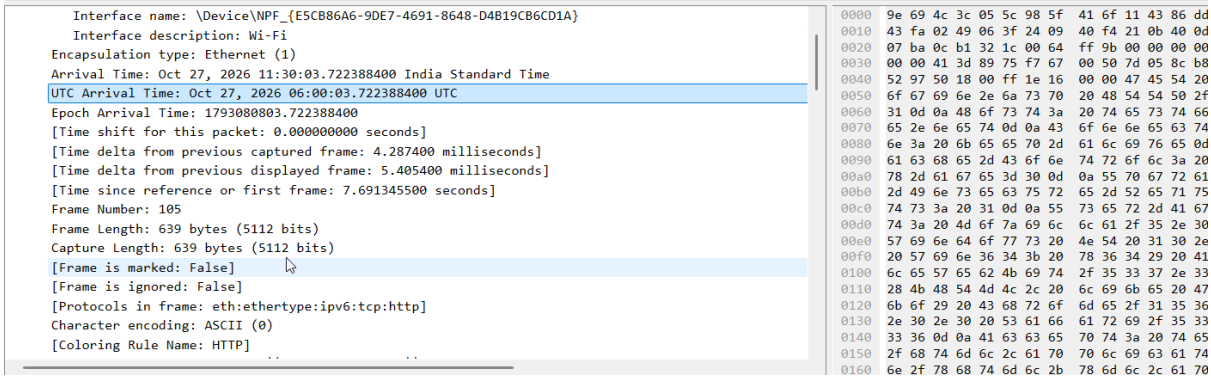
1. **Find the website's traffic.** Apply the filter http to display unencrypted web traffic.
2. **Find the login request.** Apply the filter http.request.method == "POST". The login form submission appears as a POST request.
3. **Inspect the packet.** Click the POST packet and expand **HTML Form URL Encoded** in the Packet Details pane. The username and password fields are visible in plain text.
4. **Follow the stream.** Right-click the packet and select Follow > TCP Stream to view the full request and server response.
5. **Compare with Google.** Apply the filter tls and observe that the Google traffic is encrypted and its content cannot be read.
6. **Review DNS.** Apply the filter dns to see the lookups for google.com and testfire.net.

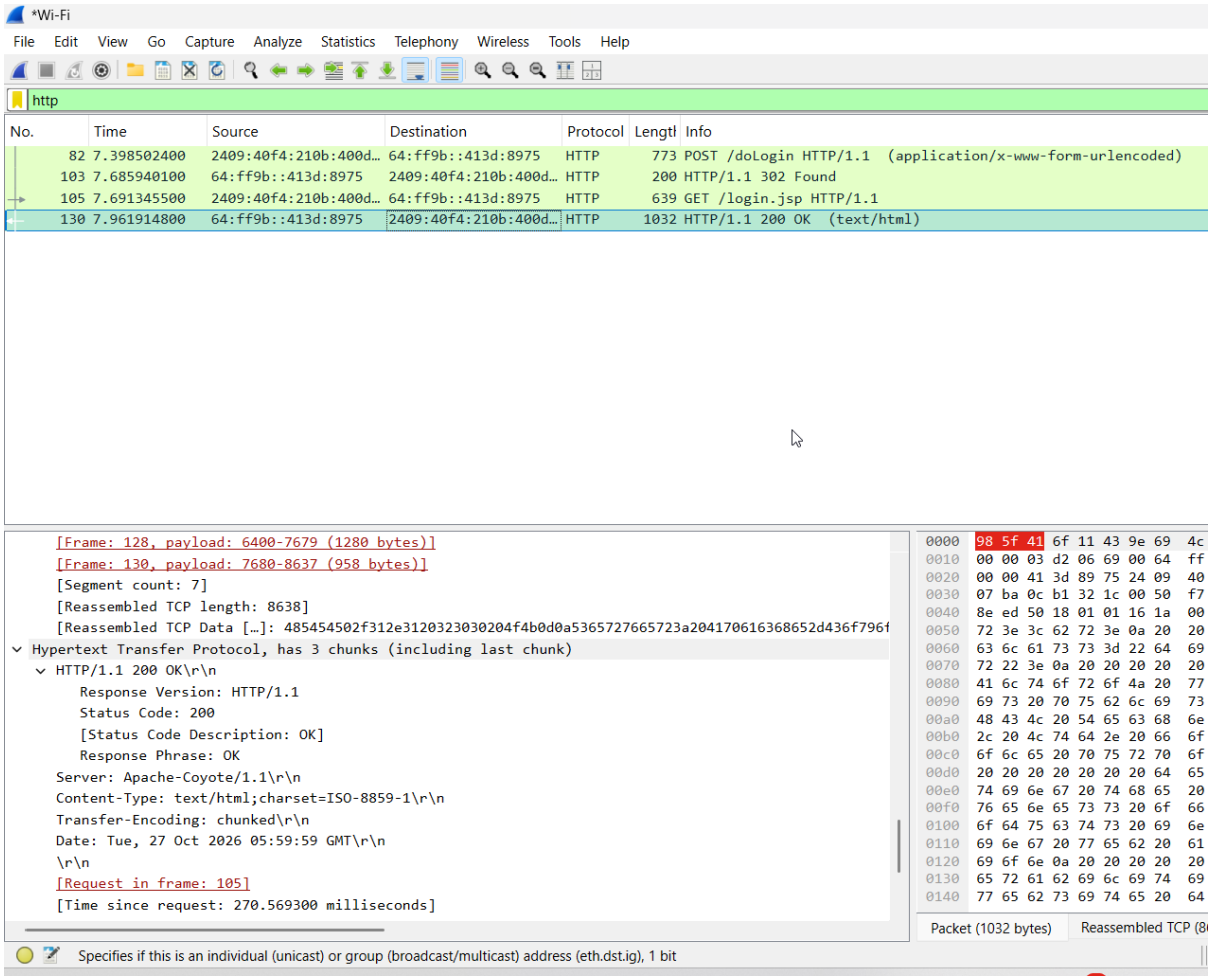



File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http
No. Time Source Destination Protocol Lengtl Info
82 7.398502400 2409:40f4:210b:400d… 64:ff9b::413d:8975 HTTP 773 POST /doLogin HTTP/1.1 (application/x-www-form-urlencoded)
103 7.685940100 64:ff9b::413d:8975 2409:40f4:210b:400d… HTTP 200 HTTP/1.1 302 Found
105 7.691345500 2409:40f4:210b:400d… 64:ff9b::413d:8975 HTTP 639 GET /login.jsp HTTP/1.1
130 7.961914800 64:ff9b::413d:8975 2409:40f4:210b:400d… HTTP 1032 HTTP/1.1 200 OK (text/html)


Upgrade-Insecure-Requests: 1\r\n
Content-Type: application/x-www-form-urlencoded\r\n
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome
Origin: http://testfire.net\r\n
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/jxl,image/avif,image/webp,image
Referer: http://testfire.net/login.jsp\r\n
Accept-Encoding: gzip, deflate\r\n
Accept-Language: en-US,en;q=0.9\r\n
Cookie: JSESSIONID=ADC45D25EAE7A368C28F8D01D8EE796A\r\n
Cookie pair: JSESSIONID=ADC45D25EAE7A368C28F8D01D8EE796A
\r\n
[Response in frame: 103]
[Full request URI: http://testfire.net/doLogin]
File Data: 37 bytes
HTML Form URL Encoded: application/x-www-form-urlencoded
Form item: "uid" = "gowsika"
Form item: "passw" = "123"
Form item: "btnSubmit" = "Login"
0000 9e 69 4c 3c 05 5c 98 5f 41 6f 11 43 86
0010 43 fa 02 cf 06 3f 24 09 40 f4 21 0b 40
0020 07 ba 0c b1 32 1c 00 64 ff 9b 00 00 00
0030 00 00 41 3d 89 75 f7 67 00 50 7d 05 89
0040 52 19 50 18 00 ff 1e 9c 00 00 50 4f 53
0050 64 6f 4c 6f 67 69 6e 20 48 54 54 50 2f
0060 0d 0a 48 6f 73 74 3a 20 74 65 73 74 66
0070 2e 6e 65 74 0d 0a 43 6f 6e 6e 65 63 74
0080 3a 20 6b 65 65 70 2d 61 6c 69 76 65 0d
0090 6e 74 65 6e 74 2d 4c 65 6e 67 74 68 3a
00a0 0d 0a 43 61 63 68 65 2d 43 6f 6e 74 72
00b0 20 6d 61 78 2d 61 67 65 3d 30 0d 0a 55
00c0 61 64 65 2d 49 6e 73 65 63 75 72 65 2d
00d0 75 65 73 74 73 3a 20 31 0d 0a 43 6f 6e
00e0 74 2d 54 79 70 65 3a 20 61 70 70 6c 69
00f0 69 6f 6e 2f 78 2d 77 77 77 2d 66 6f 72
0100 72 6c 65 6e 63 6f 64 65 64 0d 0a 55 73
0110 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c
0120 2e 30 20 28 57 69 6e 64 6f 77 73 20 4e
0130 30 2e 30 3b 20 57 69 6e 36 34 3b 20 78
0140 20 41 70 70 6c 65 57 65 62 4b 69 74 2f
0150 2e 33 36 20 28 4b 48 54 4d 4c 2c 20 6c
0160 20 47 65 63 6b 6f 29 20 43 68 72 6f 6d


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Interface name: \Device\NPF_{E5CB86A6-9DE7-4691-8648-D4B19CB6CD1A}
Interface description: Wi-Fi
Encapsulation type: Ethernet (1)
Arrival Time: Oct 27, 2026 11:30:03.722388400 India Standard Time
UTC Arrival Time: Oct 27, 2026 06:00:03.722388400 UTC
Epoch Arrival Time: 1793080803.722388400
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 4.287400 milliseconds]
[Time delta from previous displayed frame: 5.405400 milliseconds]
[Time since reference or first frame: 7.691345500 seconds]
Frame Number: 105
Frame Length: 639 bytes (5112 bits)
Capture Length: 639 bytes (5112 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp:http]
Character encoding: ASCII (0)
[Coloring Rule Name: HTTP]
0000 9e 69 4c 3c 05 5c 98 5f 41 6f 11 43 86 dd
0010 43 fa 02 49 06 3f 24 09 40 f4 21 0b 40 0d
0020 07 ba 0c b1 32 1c 00 64 ff 9b 00 00 00 00
0030 00 00 41 3d 89 75 f7 67 00 50 7d 05 8c b8
0040 52 97 50 18 00 ff 1e 16 00 00 47 45 54 20
0050 6f 67 69 6e 2e 6a 73 70 20 48 54 54 50 2f
0060 31 0d 0a 48 6f 73 74 3a 20 74 65 73 74 66
0070 65 2e 6e 65 74 0d 0a 43 6f 6e 6e 65 63 74
0080 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65 0d
0090 61 63 68 65 2d 43 6f 6e 74 72 6f 6c 3a 20
00a0 78 2d 61 67 65 3d 30 0d 0a 55 70 67 72 61
00b0 2d 49 6e 73 65 63 75 72 65 2d 52 65 71 75
00c0 74 73 3a 20 31 0d 0a 55 73 65 72 2d 41 67
00d0 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30
00e0 57 69 6e 64 6f 77 73 20 4e 54 20 31 30 2e
00f0 20 57 69 6e 36 34 3b 20 78 36 34 29 20 41
0100 6c 65 57 65 62 4b 69 74 2f 35 33 37 2e 33
0110 28 4b 48 54 4d 4c 2c 20 6c 69 6b 65 20 47
0120 6b 6f 29 20 43 68 72 6f 6d 65 2f 31 35 36
0130 2e 30 2e 30 20 53 61 66 61 72 69 2f 35 33
0140 33 36 0d 0a 41 63 63 65 70 74 3a 20 74 65
0150 2f 68 74 6d 6c 2c 61 70 70 6c 69 63 61 74
0160 6e 2f 78 68 74 6d 6c 2b 78 6d 6c 2c 61 70


*Wi-Fi
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130 7.961914800 64:ff9b::413d:8975 2409:40f4:210b:400d… HTTP 1032 HTTP/1.1 200 OK (text/html)
[Frame: 128, payload: 6400-7679 (1280 bytes)]
[Frame: 130, payload: 7680-8637 (958 bytes)]
[Segment count: 7]
[Reassembled TCP length: 8638]
[Reassembled TCP Data […]: 485454502f312e3120323030204f4b0d0a5365727665723a204170616368652d436f7961
Hypertext Transfer Protocol, has 3 chunks (including last chunk)
HTTP/1.1 200 OK\r\n
Response Version: HTTP/1.1
Status Code: 200
[Status Code Description: OK]
Response Phrase: OK
Server: Apache-Coyote/1.1\r\n
Content-Type: text/html;charset=ISO-8859-1\r\n
Transfer-Encoding: chunked\r\n
Date: Tue, 27 Oct 2026 05:59:59 GMT\r\n
\r\n
[Request in frame: 105]
[Time since request: 270.569300 milliseconds]
0000 98 5f 41 6f 11 43 9e 69 4c
0010 00 00 03 d2 06 69 00 64 ff
0020 00 00 41 3d 89 75 24 09 40
0030 07 ba 0c b1 32 1c 00 50 f7
0040 8e ed 50 18 01 01 16 1a 00
0050 72 3e 3c 62 72 3e 0a 20 20
0060 63 6c 61 73 73 3d 22 64 69
0070 72 22 3e 0a 20 20 20 20 20
0080 41 6c 74 6f 72 6f 4a 20 77
0090 69 73 20 70 75 62 6c 69 73
00a0 48 43 4c 20 54 65 63 68 6e
00b0 2c 20 4c 74 64 2e 20 66 6f
00c0 6f 6c 65 20 70 75 72 70 6f
00d0 20 20 20 20 20 20 20 64 65
00e0 74 69 6e 67 20 74 68 65 20
00f0 76 65 6e 65 73 73 20 6f 66
0100 6f 64 75 63 74 73 20 69 6e
0110 69 6e 67 20 77 65 62 20 61
0120 69 6f 6e 0a 20 20 20 20 20
0130 65 72 61 62 69 6c 69 74 69
0140 77 65 62 73 69 74 65 20 64
Packet (1032 bytes)
Reassembled TCP (8
Specifies if this is an individual (unicast) or group (broadcast/multicast) address (eth.dst.ig), 1 bit